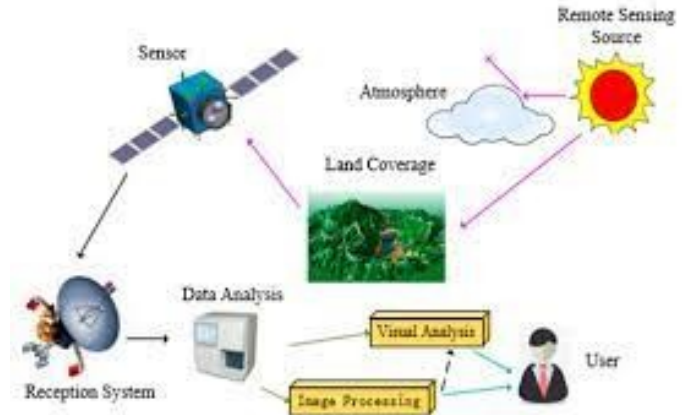


Application of Image Processing

Module 3: Satellite Image Processing

Satellite Image Processing:

Satellite image processing is the computer manipulation of digital data acquired by satellites to extract valuable information about the Earth's surface. The process is a core component of remote sensing and is crucial for extracting insights from the vast amounts of satellite imagery collected daily. In satellite image processing, imaging techniques vary by sensor and wavelength, while pre-processing standardizes the data for analysis.

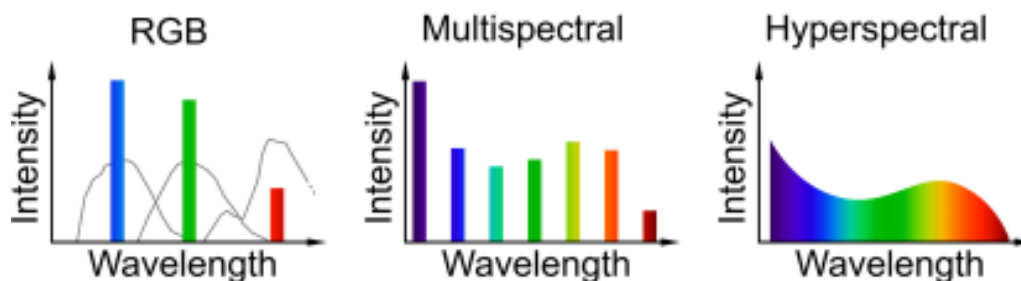
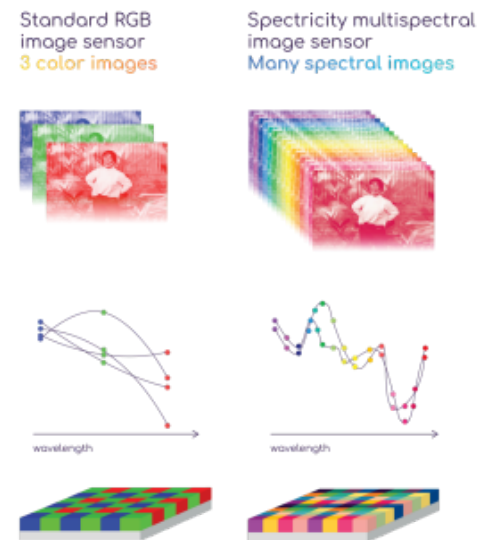


Multispectral Imaging (MSI)

Multispectral Imaging (MSI) is a technique that captures and analyzes images across multiple, specific wavelength ranges of the electromagnetic spectrum. As seen in Figure 1, compared to many off-the-shelf RGB cameras that only capture red, green, and blue wavelengths, multispectral imaging captures many spectral data points for each pixel in an image.

To give a real-world analogy, it's like seeing through a set of colored sunglasses, each tuned to reveal hidden details. For example, Spectricity's S1 Multispectral Image Sensor supports operation in 15 independent color channels, covering 400 nm to 700 nm. The richer data allows detailed analysis, identification, and advanced AI algorithm development that could not be achieved with an ordinary RGB camera.

Multi-spectral sensors capture data in a limited number of broad spectral bands, typically ranging from 3–15 bands across the visible, near-infrared (NIR), and shortwave infrared (SWIR) regions. For example, Landsat satellites acquire data in 7–11 spectral bands, while Sentinel-2 provides 13 bands.



Key concepts of Multi-spectral imaging (MSI)

- **Sensor:** Captures electromagnetic energy in a few broad, discrete bands. Examples include the Red, Green, Blue, and Near-Infrared bands, typically ranging from 5 to 20 bands.
- **Spectral resolution:** Lower spectral resolution than hyperspectral imaging due to its wider, non-continuous bands.
- **Data size:** Produces smaller, more manageable data volumes, making it faster and less resource-intensive to process.

- Applications:
 - Ideal for distinguishing general land-cover characteristics and landscape patterns.
 - Used in agriculture for crop health and in forestry for monitoring deforestation, pest and disease identification, precision irrigation, and variable rate fertilizing.
 - Calculation of vegetation indices like NDVI (Normalized Difference Vegetation Index).
 - Mapping water bodies, forests, and soil types.
 - Land cover classification at moderate spatial resolution.
 - Another application is to aid in sustainable land management by distinguishing between different crops, as well as vegetation cover and bare ground.
- Examples: Landsat and Sentinel-2 satellites.
- Advantages: Balanced between spectral richness and manageable data size.

Hyper-spectral imaging (HSI)

Unlike MSI, hyper-spectral sensors collect hundreds of contiguous narrow bands (e.g., 5–10 nm wide). This detailed spectral information creates a “spectral signature” unique to each material. Examples include NASA’s AVIRIS and ISRO’s HySIS.

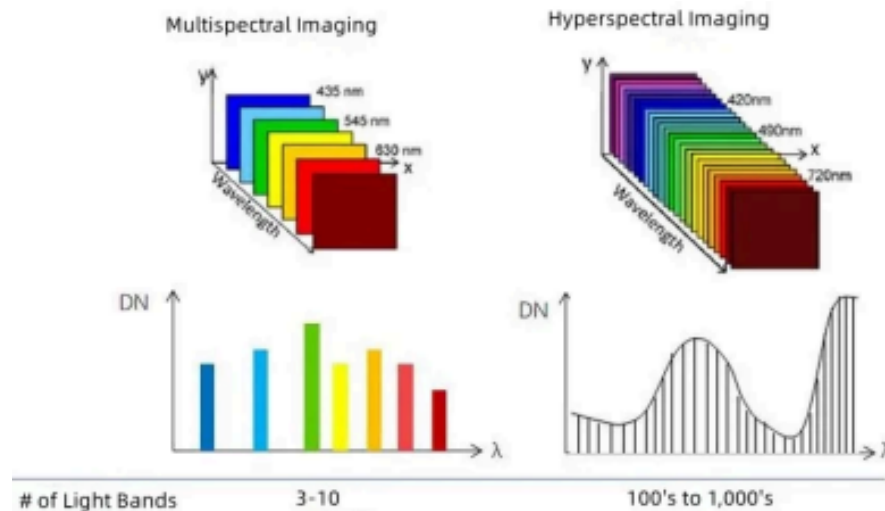
Key concepts of Hyper-spectral imaging (MSI)

- Sensor: Collects data across hundreds of very narrow, continuous spectral bands across the visible and infrared regions. This produces a continuous spectral response curve for each pixel, creating a 3D "hypercube" (two spatial dimensions and one spectral dimension).
- Spectral resolution: Higher spectral resolution, allowing for the detection of subtle variations in spectral signatures that are missed by multispectral sensors.
- Data size: Generates vast amounts of detailed data, requiring more complex processing, powerful computing, and extensive storage.
- Applications:
 - Enables precise material identification and quantification based on unique chemical and physical properties.
 - Used in geology for mineral mapping and in environmental monitoring to assess water quality and vegetation stress.
 - Identifying crop species and detecting diseases.
 - Mineral and soil composition mapping.
 - Precision agriculture and food quality monitoring.
- Examples: Hyperion and AVIRIS sensors.
- Advantages: High discrimination capability.
- Challenges: High data volume, need for advanced pre-processing. Differences in MSI and HSI

The key differences between hyperspectral and multispectral technologies are the quantity of bands captured, the narrowness of those bands (spectral resolution), and the electromagnetic radiation spectra contained within each band.

Which has more bands: multispectral or hyperspectral imaging?

- Hyperspectral imaging (HSI), as the name suggests, covers a larger number of spectral bands than multispectral imaging (MSI).
- Multispectral satellite constellations collect data throughout a limited portion of the electromagnetic spectrum (five to ten bands); typically, it’s RGB (red, green, and blue — the three primary colors) and a handful of infrared bands. Hyperspectral satellites can distinguish thousands of individual light spectrum bands.



Here we see an example where MSI includes quite wide short-wave infrared, near-infrared, and RGB bands. Numerous extremely narrow bands (10–20 nm), which do not have descriptive names, make up hyperspectral images.

Both hyperspectral and multispectral imaging satellites mainly operate in LEO (low Earth orbit). MSI satellites, such as Landsat, Sentinel-2, and EOS SAT-1, provide information essential for distinguishing land cover characteristics and landscape patterns. Hyperspectral sensors see substantially more information, so they can even identify and quantify materials. While extra bands do make it easier to distinguish more details, they also require the elimination of redundant data and make the analysis more complex and expensive.

Spectral And Spatial Resolutions: HSI offers a high spectral resolution as it can distinguish between wavelengths precisely. At the same time, it is associated with low spatial resolution. Multispectral sensors, as opposed to hyperspectral, are more suited to applications that need general spectral data since they operate within broader and fewer bands. Having said that, the spatial resolution of MSI is usually higher.

Data Size And Processing: Due to the vast amount of detailed data produced by hyperspectral remote sensing, data processing requires complex and resource-intensive methods. For proper satellite imagery interpretation, you'll need specialist software and knowledge.

Cost: The complexity of the sensors and the subsequent processing of the acquired data define the pricing of either option. When compared to multispectral, hyperspectral imaging is often more costly and resource-intensive. The former, a more affordable option, is used unless the exact data, such as on materials, is needed.

Dependence On Atmospheric Conditions: Hyperspectral vs. multispectral remote sensing is easily affected by environmental conditions and requires thorough calibration. Because of this, HSI should only be used in controlled settings or for very particular scientific studies. Multispectral imaging does not have these restrictions; it works well in a wide range of environments and is less impacted by atmospheric interference, hence it has more potential uses.

HSI has a lot of untapped potential, but for the time being, it is primarily employed in scientific research. On the other hand, MSI is more than enough for the needs of casual users across a range of remote sensing applications. Most satellites and constellations offer multispectral satellite imagery in real-time that is readily available on the EOSDA LandViewer platform.

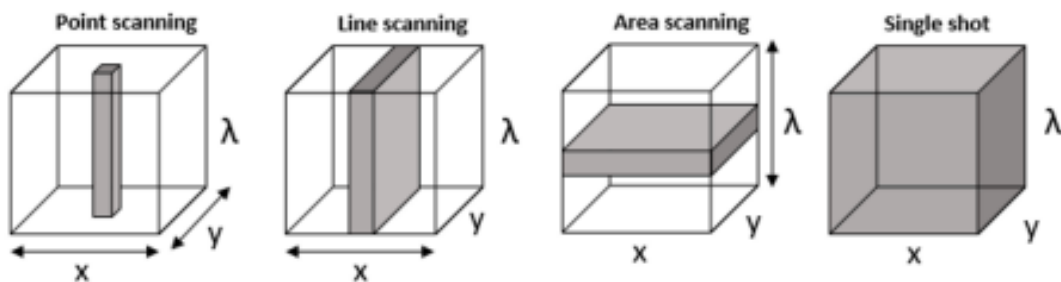
Compared to hyperspectral, multispectral data is easier and faster to process because there are fewer spectral bands. This

works well in situations where quick or real-time analysis is of the utmost importance. Nowadays, the analysis of MSI is made easier with widely open, user-friendly software and resources, making the technology accessible to a variety of industries and applications.

The acquisition system employed in HSI/MSI has three main characteristics: spectral range, spectral resolution, and spatial resolution.

1. The spectral range is the range of wavelengths covered by the spectral camera: the visible and near infrared (VNIR) spectrum (400 to 1000 nm), the NIR spectrum (from 900 to 1700 nm), or the short wavelength infrared (SWIR) spectrum (from 1000 to 2500 nm) are common spectral ranges.
2. The spectral resolution is the difference between consecutive wavelengths.
3. The spatial resolution is linked to the actual pixel size, which depends on the number of pixels in the x and y directions as well as the size of the image taken by the camera. These three characteristics will have to be chosen depending on the final application.

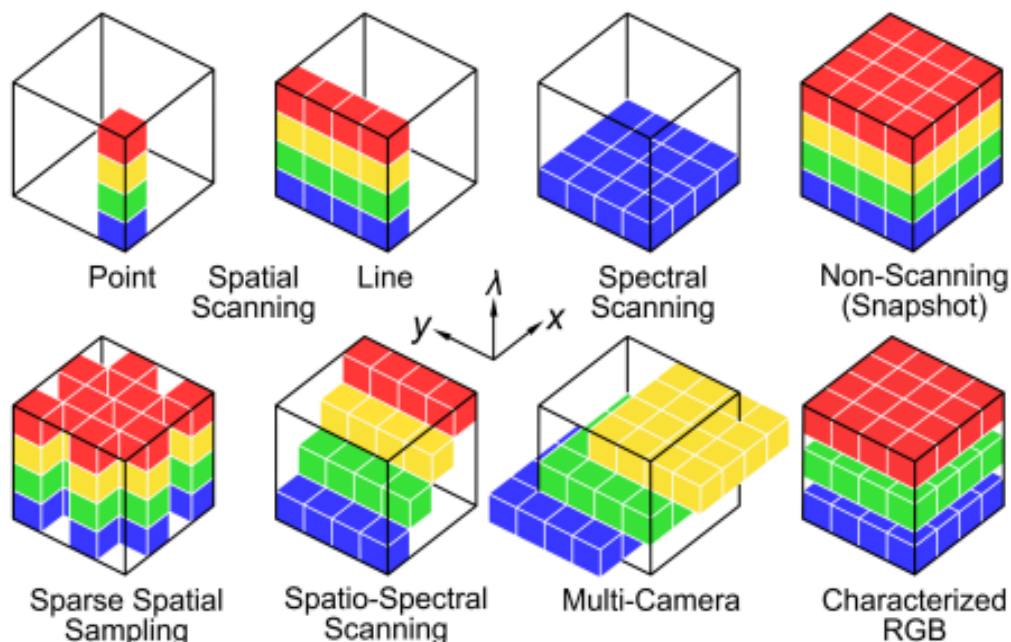
Different types of acquisition systems can be distinguished depending on how the spectral (λ) and the spatial (x,y) information is obtained.



Spatial scanning techniques consist of obtaining the spectrum for a single point (point scanning or whiskbroom mode) or a single line (line scanning or pushbroom mode) and doing a spatial sweep (in (x,y) for point scanning and in x for line scanning).

Spectral scanning methods consist of obtaining the spectrum for all the space (x,y) and a single wavelength, and doing a spectral sweep in λ . This method is also known as plane scanning or area scanning mode.

A single-shot imager or snapshot sensor acquires both the spatial (x,y) and the spectral (λ) information on a single shot.



Applications Of Multispectral And Hyperspectral Technologies

Multispectral and hyperspectral imaging have numerous practical uses in different areas, including farming, environmental research, disaster management, and more. Whereas hyperspectral images are more commonly used, for example, in geology and mineral development to gather information about materials, MSI is used in agriculture and forestry to gather data about the earth's surface, its cover, and patterns of change.

Precision Agriculture

Wrt accessibility and ease of analysis compared to hyperspectral data, multispectral has become common in precision farming solutions for agricultural consultants and other agribusiness players. Fine-resolution MSI supports farmers in crop health monitoring, pest and disease identification, precision irrigation, and variable rate fertilizing. Another application is to aid in sustainable land management by distinguishing between different crops as well as vegetation cover and bare ground.

Vegetation Analysis

Many aspects of vegetation cover are studied using multispectral and hyperspectral data. Vegetation change detection, particularly using MSI, is essential for monitoring how plant cover in a certain region changes over time. Satellite imaging in agriculture may also be used to assess plant diversity as a part of biodiversity research.

Environmental Monitoring

MSI is perfect for spotting changes in land use and plant cover and assessing the health state of different ecosystems, such as forests and wetlands. When mapping hydrological structures, hyperspectral images could be helpful. Researchers are increasingly turning to hyperspectral and multispectral remote sensing to better monitor bodies of water, their pollution, and water quality changes.

Disaster Management

There is a strong reliance on remote sensing technologies for emergency preparedness and disaster assessment. MSI is highly beneficial in estimating the damage caused by earthquakes, floods, hurricanes, and other natural and man-made calamities. On top of that, by delivering data in near-real time from even the most remote and undeveloped regions, it aids in disaster response coordination and allocation. Multispectral data, for instance, was extensively employed for analyzing and mitigating natural disasters in 2023.

Forestry

Advancements in multispectral and hyperspectral sensors have greatly simplified and improved the accuracy of remote tree species detection. When it comes to evaluating forest health, keeping tabs on deforestation, and sustainable forestry management, MSI is invaluable. By closely monitoring changes and deviations in plant health, spectral imaging plays an important part in detecting regions that may be at risk of wildfires.

Thermal infrared (TIR) imaging

- Sensor: Detects the thermal infrared energy naturally emitted from objects with a temperature greater than absolute zero.
- Data: Different objects radiate different amounts of thermal energy, which can be used to distinguish between materials like vegetation, water, and soil.
- Applications: Used to measure and map temperature variations. Applications include monitoring weather patterns, tracking lava flows, detecting forest fires, and identifying geothermal anomalies.
- Examples: Thermal sensors on satellites like NASA's Landsat series.

Radar imaging

- Sensor: An active remote sensing system that transmits its own microwave energy and measures the energy backscattered by the Earth's surface.
- Advantages: Not dependent on solar illumination and can penetrate cloud cover and rain, allowing for all-weather, day-or-night imaging. It is also sensitive to the physical structure and moisture content of surfaces.
- Applications: Used for digital elevation mapping, vegetation and soil moisture mapping, geological mapping, urban planning, and disaster management.
- Examples: Synthetic Aperture Radar (SAR) systems on satellites like RADARSAT and TerraSAR-X.

Radiometric correction

Purpose: Adjusts the raw digital numbers (DNs) of an image to account for sensor errors and variations in atmospheric and illumination conditions. This process converts DN values into standardized, physically meaningful units like radiance or reflectance, enabling quantitative analysis and comparison between different images.

Methods:

- Sensor calibration: Corrects for inconsistencies in the sensor's response to radiation.
- Atmospheric correction: Removes the effects of atmospheric scattering and absorption caused by particles and gases. Common techniques include Dark Object Subtraction and using atmospheric modeling software.
- Noise reduction: Removes random "noise" and systematic errors like striping or banding caused by sensor malfunction.

Geometric correction

Purpose: Corrects for spatial distortions that alter the true location of image pixels and aligns the image with a standard map projection. This ensures accurate measurements of distance, area, and direction and allows images from different dates to be compared.

Methods:

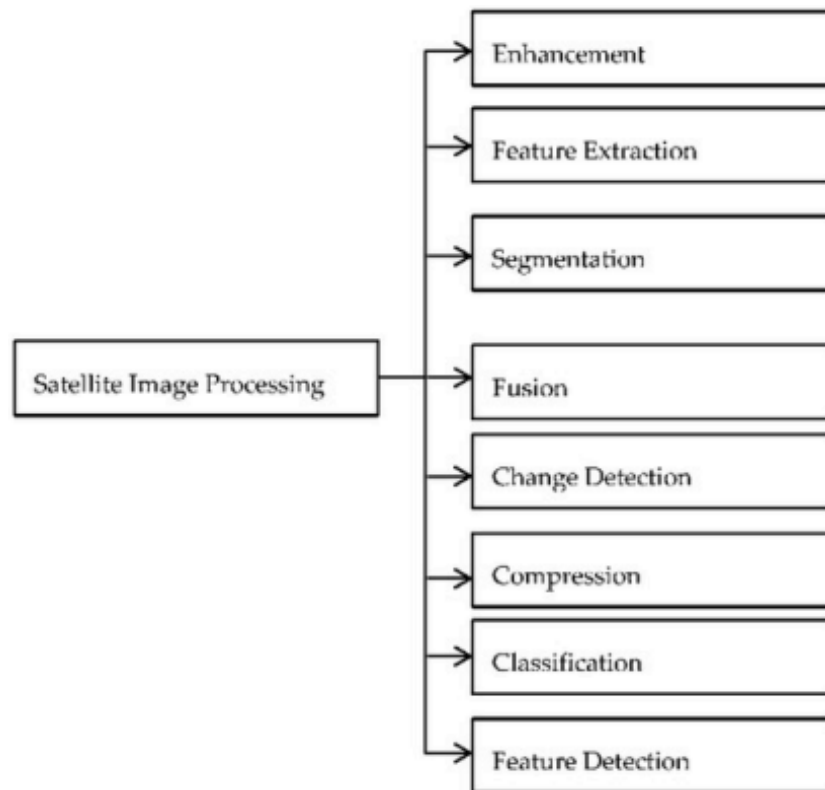
- Image-to-map registration: Uses ground control points (GCPs), which are easily identifiable points with known real-world coordinates, to establish a mathematical relationship and transform the image.
- Orthorectification: Removes geometric distortions caused by terrain relief using a Digital Elevation Model (DEM).
- Resampling: Assigns new brightness values to pixels in the geometrically corrected output image. Common methods include Nearest Neighbor, Bilinear Interpolation, and Cubic Convolution.

Pre-processing of satellite images

The key stages of satellite image processing are:

- Preprocessing: Correcting raw satellite data to remove errors and distortions.
- Enhancement: Improving the visual quality of the image to make features more visible.
- Analysis and Feature Extraction: Identifying specific elements like roads, buildings from the images.
- Classification: Categorizing pixels into distinct groups or land-cover types

Pre-processing is the initial phase of satellite image processing, which corrects raw image data for distortions caused by sensor, atmospheric, and terrain factors.



Data Acquisition and Level 0 Processing (L0)

This is the initial step where raw data is received directly from satellite sensors by ground tracking stations. The raw bits of data represent radiance values captured by sensors but are not yet processed for errors or distortions. Level 0 data contains no corrections and is usually handled by satellite agencies before public distribution.

Scene Extraction

Satellites capture data in long strips or swaths along their orbital paths. These strips are segmented into individual "scenes," which are manageable image units that typically cover specific geographic extents. Scene extraction separates these along-track strips into frames to facilitate further processing.

Radiometric Correction

Radiometric correction addresses sensor irregularities, converts raw pixel digital numbers (DNs) to physical units like reflectance or radiance, and compensates for atmospheric interference and sensor noise. This step enables accurate quantitative analysis and comparability across images acquired at different times or conditions. It involves:

- Calibration with sensor parameters and metadata,
- Conversion of DN to at-sensor radiance,
- Atmospheric correction models (e.g., 5S models),
- Correction for solar angle and Earth-Sun distance variations,
- Removal of sensor noise such as striping and dead pixels.

The goal is to represent the true surface radiance values from the Earth's surface.

Geometric Correction

This step corrects geometric distortions caused by satellite sensor viewing angles, Earth curvature, satellite platform motion, and terrain relief. It involves:

- Georeferencing pixels to real-world geographic coordinates,
- Rectifying images to map projections or coordinate systems,

- Using ground control points (GCPs) or digital elevation models (DEMs) to improve spatial accuracy. Geometric correction ensures images can be accurately overlaid with other spatial data for analysis or map creation.

Noise Reduction / Speckle Filtering (especially for SAR)

Synthetic Aperture Radar (SAR) images suffer from speckle noise—a granular interference disrupting visual clarity and analysis. This step uses filtering algorithms to:

- Reduce speckle while preserving edges and important features,
 - Enhance interpretability and feature extraction.
- Filters such as Lee, Frost, or Gamma filters are common.

Terrain Correction (specifically for Radar)

Radar images distort when radar waves reflect off terrain slopes and vary with elevation, causing geometric displacement (layover, foreshortening). Terrain correction compensates by incorporating DEMs to:

- Correct spatial distortions,
- Improve the positional accuracy of radar backscatter values.

Image Enhancement

To improve visual interpretability and support human or algorithmic analysis, images are enhanced by:

- Adjusting brightness and contrast,
 - Applying histogram equalization,
 - Using sharpening filters or edge enhancement.
- These techniques make subtle features more distinguishable.

Image Stitching / Mosaicking

To analyze large areas, multiple adjacent scenes are combined into a seamless mosaic by:

- Aligning and merging images with geometric correction to avoid gaps or overlaps,
 - Balancing brightness and colors for uniform appearance.
- This produces wider area coverage from several scenes.

Cloud Masking and Atmospheric Correction (optical images)

Clouds and atmospheric particles distort optical satellite images by obscuring the surface. This step:

- Detects and removes cloud-affected pixels,
- Applies atmospheric correction models to adjust for haze, aerosols, and scattering effects,
- Restores accurate surface reflectance values for land cover analysis.

Resampling and Reprojection

After geometric correction, data may be resampled to a different pixel size or spatial resolution to standardize datasets. Images are also reprojected into coordinate systems matching other geospatial layers for integrated analysis.

Satellite image processing

Satellite image processing deals with analyzing digital images captured by artificial satellites to gather coherent information about the Earth's surface for different scientific and practical applications such as environmental monitoring, land use planning, disaster detection, and research.

Resolutions in Satellite Images

- Spatial resolution: Pixel size on the ground; detail visible in the image.
- Spectral resolution: Number of wavelength intervals the sensor can detect.
- Temporal resolution: Frequency of image capture over the same area.
- Radiometric resolution: Bit depth indicating the levels of brightness recorded.

Image Processing Categories

1. Image Rectification and Restoration: Correcting geometric distortions, calibrating radiometric data, and removing noise.
2. Image Enhancement: Improving the visual quality and distinction between features (e.g., contrast adjustment).
3. Information Extraction: Automated classification and feature identification using quantitative techniques and algorithms.

Preprocessing Steps

- Level 0 Processing (L0): Initial reception and basic conversion of raw satellite data.
- Level 1 Processing (L1): Includes scene extraction (dividing data into manageable images), radiometric correction to clean image data, and geometric correction to correct spatial distortions.
- Radiometric Correction: Adjustments for sensor variations and atmospheric effects.
- Geometric Correction: Aligns images to a map coordinate system or other reference images for proper spatial analysis.

Common Image Processing Techniques

- Image Enhancement (contrast, filtering)
- Image Classification (assigning land cover types)
- Change Detection (observing differences over time)
- Image Fusion (combining images for better information)
- Vegetation Index Calculation
- Feature Extraction
- Data Visualization.

Applications: Useful in agriculture, forestry, urban planning, environmental monitoring, disaster management, and scientific research. Machine learning and AI techniques are increasingly applied to satellite imagery for object detection, segmentation, and classification.

Land Use / Land Cover (LULC) Change Analysis

- LULC change analysis detects how the Earth's surface changes over time due to natural or human activities such as urban expansion, deforestation, and agricultural land conversion.
- Satellite remote sensing combined with GIS is extensively used to monitor, map, and quantify these changes across large areas and multiple years by utilizing multi-temporal satellite images like Landsat, Sentinel-2 etc.
- Post-classification comparison is a common change detection technique where classified land cover maps from different years are compared to detect transitions between land use classes.
- Machine learning algorithms (e.g., Random Forest, SVM, XGBoost) improve the accuracy of land cover classification from satellite imagery.
- Studies have shown that built-up areas and barren lands tend to expand while croplands may decline in some regions, emphasizing the need for sustainable land management policies.
- LULC change analysis helps stakeholders in environmental planning, biodiversity conservation, resource management, and policy-making by providing accurate spatial and temporal information on landscape dynamics.

Crop Type Mapping and Monitoring in Agriculture

- Crop type maps identify the spatial distribution of specific crops, which is crucial for agricultural monitoring applications including early warning of crop shortfalls, condition assessments, production forecasts, damage assessments, insurance, and climate policy.
- Global-scale, up-to-date, and harmonized crop type maps are still lacking but recent efforts by GEOGLAM (Group on Earth Observations Global Agriculture Monitoring) have developed the first unified global crop-specific maps for wheat, maize, rice, and soybeans.
- These maps are derived by harmonizing multiple national and regional satellite-derived datasets combined with expert knowledge and crop statistics.
- The maps are continuously updated and support the monitoring of crop conditions through the growing season using satellite vegetation indices such as NDVI.
- Accurate crop type mapping relies on moderate to high spatial resolution satellite data (10-250 m), as finer resolution improves characterization of fields especially in regions with complex landscapes and smallholder farms.
- Challenges include the lack of ground reference data for calibration and validation, multiple cropping seasons in a year, and timely availability of within-season crop maps.
- Crop phenology and crop calendars complement satellite data to enable monitoring of crop development stages, yield forecasting, and responses to extreme weather events.

Common applications

- Agriculture: Monitoring crop health, forecasting yields, and analyzing environmental impacts.
- Urban planning: Analyzing land use patterns and tracking urban sprawl.
- Environmental monitoring: Assessing the extent of deforestation, tracking water quality, and monitoring changes in polar ice.
- Disaster management: Assessing damage from floods, wildfires, or other natural hazards.
- Meteorology and oceanography: Tracking weather patterns, storm formations, and monitoring sea surface temperatures.